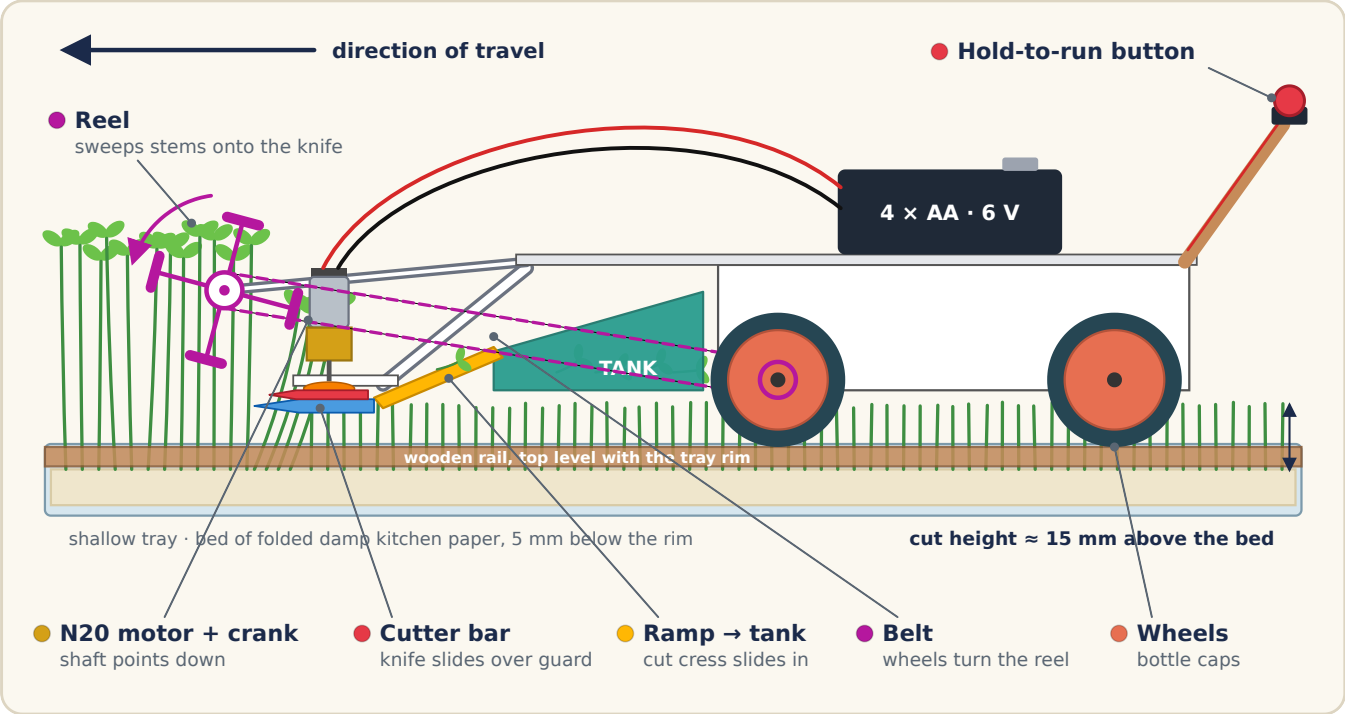
JW PROJECTS · REPORT

Cress Farming

Building a cress combine harvester with an N20-motor cutter bar — design, step-by-step build guide, safety plan and upgrade manual.



Companion to the interactive presentation at brendanjameslynskey.github.io/Cress_Farming · September 2026

Summary for parents

What it is. A small working model harvester, about 24 cm long, built mostly from foam board, lolly sticks, bamboo skewers and bottle caps. It rolls along two wooden rails beside a shallow tray of home-grown cress. A paddle reel sweeps the cress onto a plastic cutter bar, which snips it with a scissor action, and the cut cress goes up a short ramp into a small tank.

How the cutter works. A 6 V, 300 RPM “N20” gear motor — a standard hobby motor about the size of a sugar cube — turns a small crank. A lolly-stick arm turns that rotation into a 10 mm back-and-forth slide, moving a plastic knife over a fixed plastic guard comb. That makes ten gentle scissor cuts a second. It is the same principle as the knife on a real combine harvester.

Safety at a glance

6 V AA batteries (no shock risk); plastic knife; self-resetting fuse; ON/OFF switch; hold-to-run button; batteries removed before touching the cutter; an adult does all craft-knife and hot-glue work.

Cost and time

About £20 if household items are used (about £28 all new). Two afternoons of building; the cress grows for a week in parallel. Everything fits in a shoebox.

What gets learned

Plant biology, measuring, ratios and rates, mechanisms (cranks and gearboxes), simple circuits and safety devices, and methodical testing and fault-finding.

A DESIGN, NOT A TESTED KIT

The dimensions in this report are careful starting points. Chapter 19 (Troubleshooting) lists what to adjust if something doesn’t work first time, which is a normal and valuable part of an engineering project.

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Part 1 · The idea

1 The pitch — why this is a brilliant idea

Mum, this project **grows real food and builds a real machine** at the same time. The cress takes a week to grow on the windowsill. While it grows, I build a small harvester, about the size of a shoebox lid, that rolls along two wooden rails beside the tray. It snips the cress with a knife that works like a pair of scissors, and sweeps it into a little tank.

COST	BUILD TIME	GROW TIME	BATTERY	KNIFE
≈ £20	2 afternoons	7 days	6 V	Plastic

It’s safe

Four AA batteries — the same as a toy car, far too low to give a shock. The knife is **plastic**, not metal. It only runs while a button is **held down**, and a self-resetting fuse cuts the power if anything jams.

It’s cheap and tidy

Foam board, lolly sticks, bottle caps, skewers and one £4 motor. It all fits in a shoebox. The mess is kitchen paper and a few seeds. No paint, no sawdust.

It’s real engineering

The cutter works exactly like the knife on a real combine harvester. Building it means measuring, working out gear ratios, wiring a circuit and fixing things when they don’t work first time.

It’s lunch

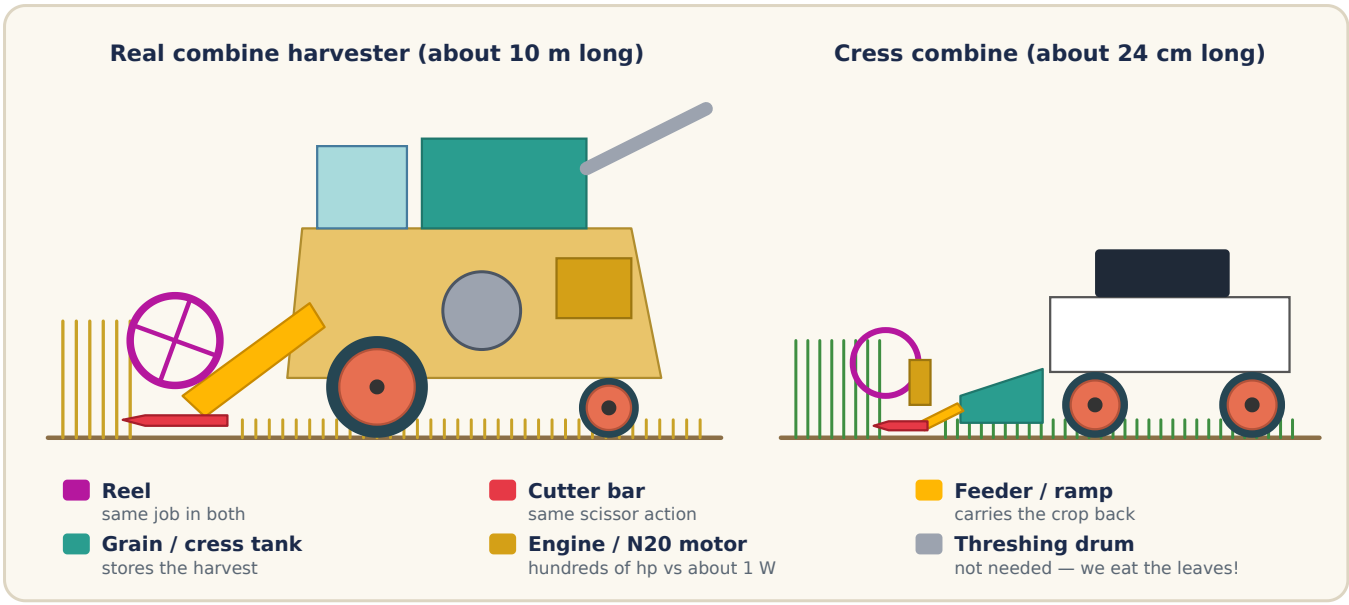
Cress is one of the easiest things anyone can grow — and at the end there are egg & cress sandwiches for everyone. I grew them, harvested them and made them.

THE DEAL

I do the planning, measuring, building, wiring and testing. A grown-up does the **craft-knife cutting** and the **hot glue**, and goes through the safety checklist with me before the first run. Everything is written down in this presentation and in the PDF report.

2 How a real combine harvester works

A combine harvester is called a *combine* because it combines several farm jobs into one machine. Driving across a wheat field, it does five things in a row:



Same colours, same jobs. The cress combine keeps the reel, the cutter bar, the feeder and the tank — and leaves out the threshing drum, because we eat the whole leaf, not just the seeds.

1 · Reel

A big slowly turning paddle wheel at the front. It gently pushes the stalks back onto the knife so they fall the right way.

2 · Cutter bar

A long row of pointed guards with a knife sliding back and forth over them. Every stalk gets snipped like scissors, about **1,000 times a minute**.

3 · Feeder

An auger and a conveyor carry the cut crop back and up into the machine.

4 · Threshing

A spinning drum knocks the grain out of the ears; sieves and a fan blow the chaff away. **We skip this** — cress leaves are the bit we want.

5 · Grain tank

Clean grain collects in a tank on top, and a long pipe (the unloading auger) empties it into a trailer.

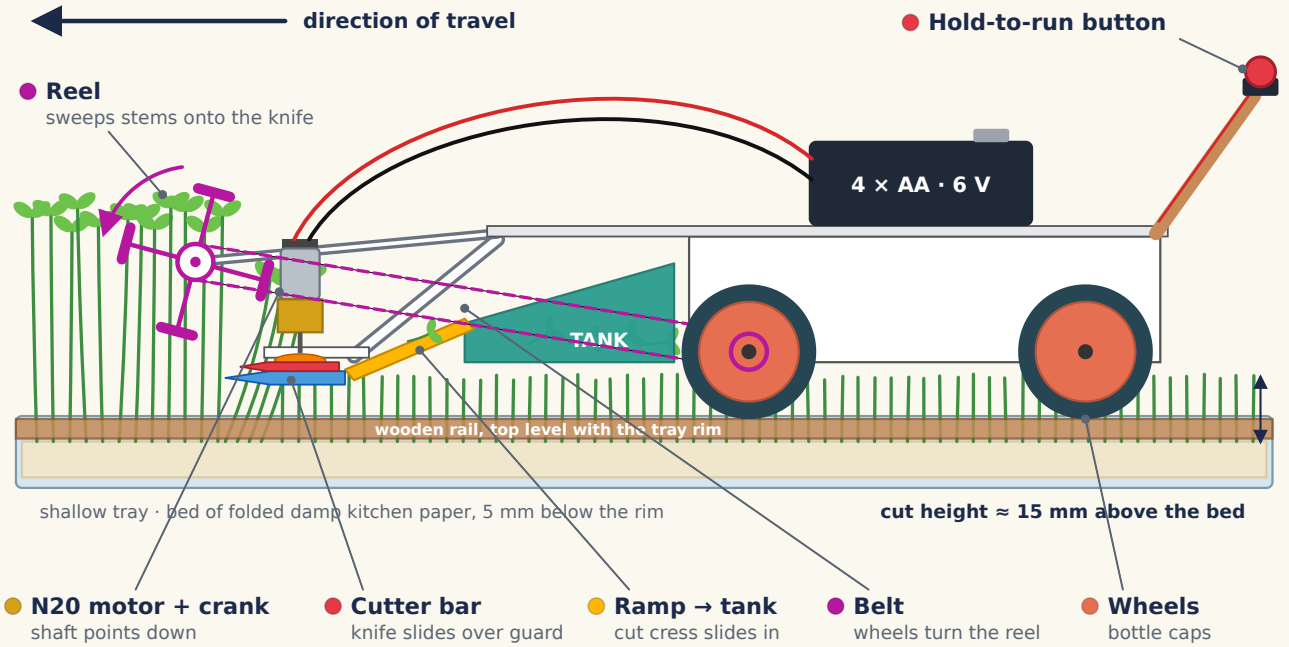
And the cress combine?

Reel, cutter bar, a ramp instead of the feeder, and a tank. Four of the five jobs, at about one-fortieth of the size.

	Real combine	Cress combine
Length	About 10 m	About 24 cm
Cutting width	6–12 m	90 mm
Power	Hundreds of horsepower (a diesel engine)	About 1 watt (an N20 motor)
Knife cuts per minute	About 1,000	About 600
Price	Hundreds of thousands of pounds	About £20
Harvest	Wheat, barley, oilseed rape	Cress (and maybe mustard)

Part 2 · The design

3 Meet the Cress Combine



Side view. The machine rolls on bottle-cap wheels along two wooden rails either side of a shallow cress tray. The reel sweeps, the knife snips, and the cut cress slides up a short ramp into the tank.

HOW IT WORKS IN ONE SENTENCE

You push it along the rails holding the button; the wheels turn the reel through a rubber-band belt; the reel sweeps cress onto a knife that snips back and forth **ten times a second**; and the cut cress is pushed up a ramp into the tank.

Reel

Four card paddles on a skewer. Driven by the wheels, so it only turns when the machine moves.

Cutter bar

A plastic guard comb (fixed) with a plastic knife (sliding) on top — ten teeth, 10 mm apart.

N20 motor + crank

A tiny geared motor turns a crank; a lolly-stick arm turns the spinning into sliding.

Ramp & tank

A gentle card ramp behind the knife leads into a dustpan-shaped plastic tub.

Battery & button

4 × AA box on top; fuse, ON/OFF switch and a hold-to-run button on the handle.

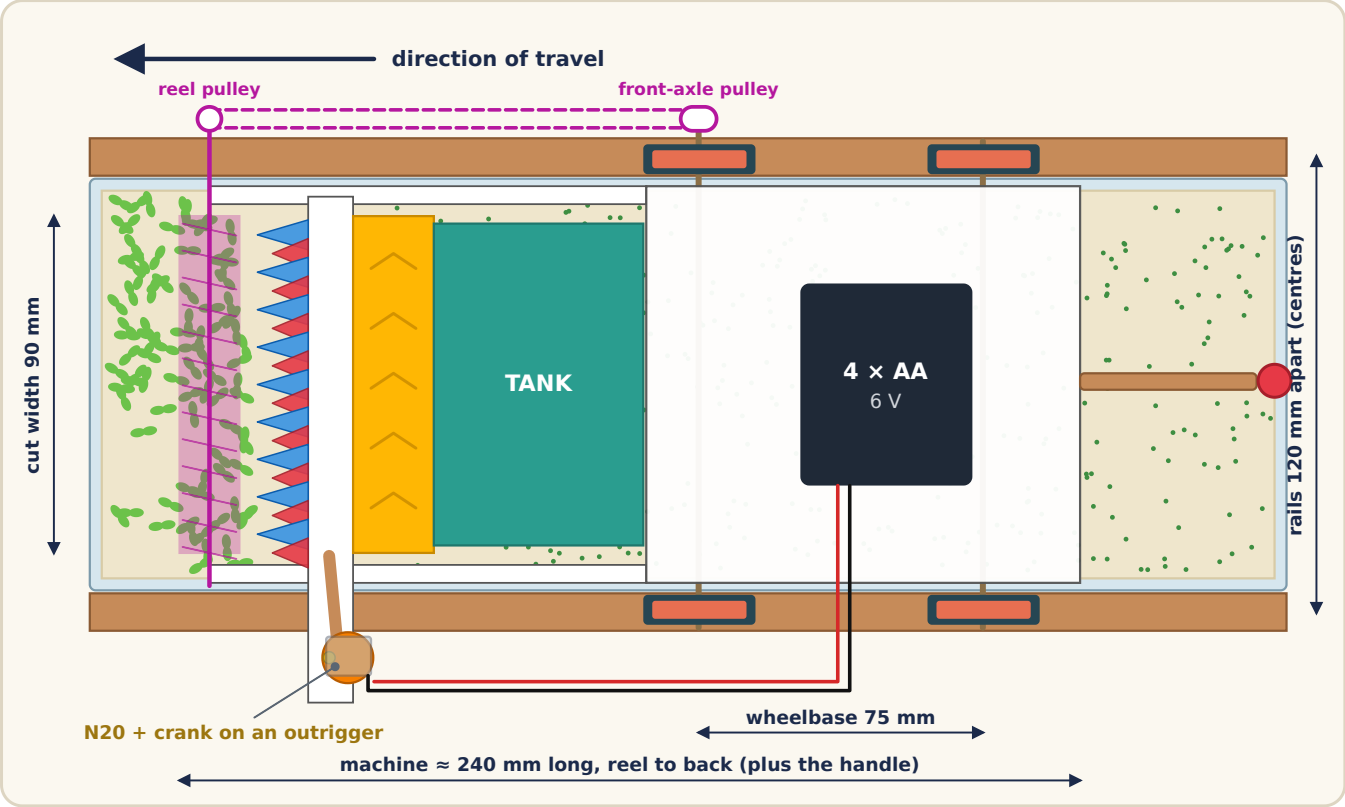
Chassis & rails

Foam-board body, skewer axles in straw bearings, bottle-cap wheels on 10 mm wooden rails.

HONEST NOTE

This is a **design to build**, not a kit that has been tested a hundred times. Every size here is a sensible starting point, and the troubleshooting page says what to change if something doesn’t work first time. That’s normal — it’s what engineers do.

4 Top view and key sizes



Looking down from above. The motor sits on an outrigger beyond one rail, so it never touches the crop. The belt runs down the other side from the reel pulley to the front axle.

Size	Value	Why this number
Cut width	90 mm (100 mm guard)	Fits a 110 mm-wide tray between the rails
Rail spacing	120 mm, centre to centre	10 mm rails sit right against the tray walls
Tray	≤ 110 mm wide, 20–30 mm deep	Shallow, so the cutter passes just above the rim
Cress bed	About 5 mm below the rim	Build it up with folded kitchen paper
Cut height	15 mm above the bed	Leaves the roots and seed cases behind
Wheels	30 mm bottle caps	Radius 15 mm — sets the reel speed ratio
Wheelbase	75 mm	Long enough to run straight, short enough to be light
Tooth pitch / stroke	10 mm / 10 mm	Each knife tooth sweeps exactly one gap

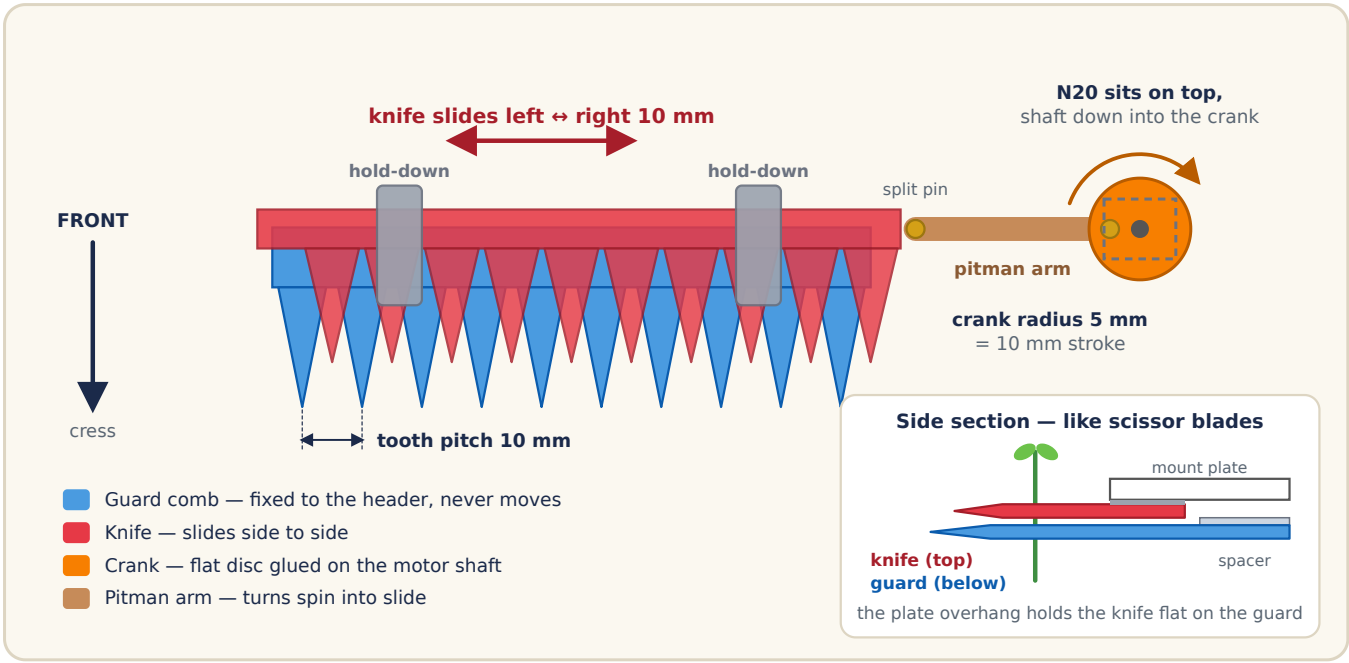
WIDER TRAY?

If your tray is wider than 110 mm, move the rails out and make the cutter parts longer. Upgrade 5 shows how, and how to check the motor can cope.

5 The cutter: an N20 motor and a pair of scissors

An **N20 gear motor** is a tiny electric motor joined to a gearbox. The whole thing is about 12 mm across and 25 mm long — smaller than a sugar cube and a half — and weighs about 10 g. The gearbox trades

speed for strength: the motor inside spins thousands of times a minute, but the output shaft turns slowly and strongly. We use the **6 V, 300 RPM** version: five turns a second.



The cutter bar from above. The motor sits on top of a crank disc. A pin 5 mm from the centre drives the pitman arm, which pushes and pulls the knife 10 mm left and right. Inset: the knife sandwiched flat between the guard and the mount plate.

Turning spin into slide

A pin on a spinning disc goes round in a circle. Join a stick to the pin and the other end moves back and forth — exactly how a steam locomotive's connecting rods work, but backwards. On a combine this stick is called the **pitman arm**.

Why not a spinning blade?

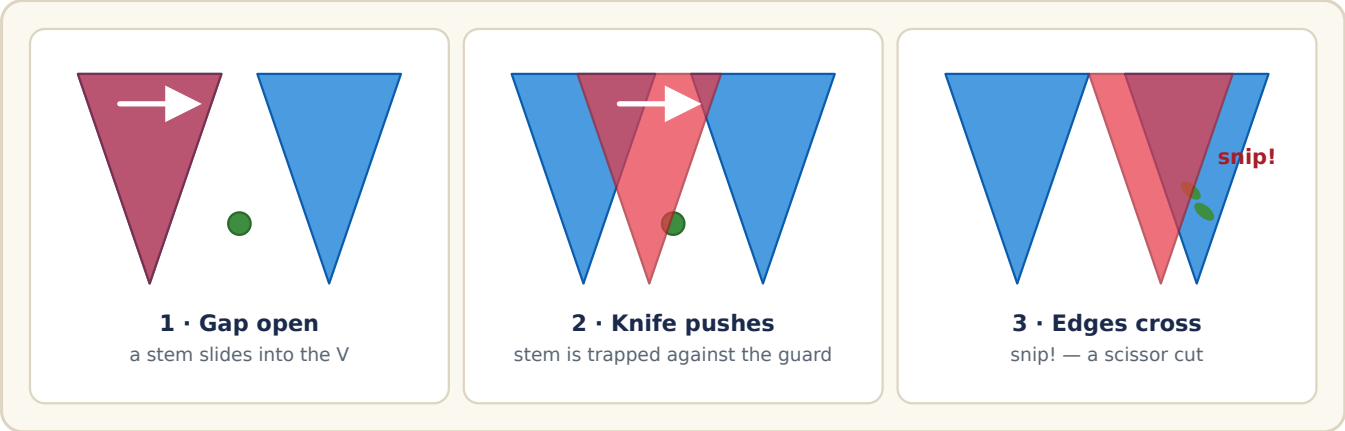
A spinning blade has to go very fast to cut soft stems — think of a strimmer — and a fast blade is dangerous. A scissor action cuts soft cress **slowly and gently**, using plastic teeth. It is the safer design, and it's also what real combines use.

STROKE = PITCH

The pin is 5 mm from the centre, so the knife travels 10 mm each way — exactly the gap between two teeth. Every knife tooth sweeps across one whole gap on every half-turn, so no stem can hide between teeth.

6 How a scissor cut works

Scissors don't cut because they are sharp — they cut because two edges **slide past each other while pressed together**. Anything caught between them is pinched and sheared. That's why the knife and guard must lie flat against each other; if there's a gap, a soft stem just folds into it.



Looking down on two blue guard teeth and one red knife tooth. The V-shaped gaps funnel each stem to the point where the edges cross.

The V funnels

Pointed teeth guide stems into the gaps as the machine moves forward, so each one is cut close to where the edges meet — the strongest part of the cut.

Flat and snug

The mount plate overhangs the knife and holds it flat on the guard. Tight enough not to lift; loose enough to slide with a light push.

Guards stick out further

Guard teeth are 15 mm long, knife teeth only 12 mm. The fixed guard always reaches further forward than the moving knife.

7 The numbers behind a clean cut

There are only three numbers to understand. The rest is just building carefully.

Number	Sum	With our design
Cuts per second	2 × (motor RPM ÷ 60) — one cut each way	2 × (300 ÷ 60) = **10 cuts/s**
Distance per cut	push speed ÷ cuts per second	20 mm/s ÷ 10 = **2 mm**
Reel speed ratio	reel radius ÷ wheel radius	25 ÷ 15 ≈ **1.7**

The golden rule

Between two cuts the machine should move forward **less than half a tooth gap (5 mm)**. Then every stem is caught by a V before the next stem arrives, and nothing gets flattened.

Distance per cut	What happens
≤ 5 mm	Clean cut — every stem snipped
5–10 mm	Mostly cut — a few stems bend over
> 10 mm	Pushing too fast — stems get flattened, not cut

SO HOW FAST DO I PUSH?

With a 300 RPM motor the knife makes 10 cuts a second, so any push speed up to **50 mm a second** keeps within the golden rule. A comfortable 20 mm a second — a 20 cm tray in 10 seconds — gives lots of margin.

Part 3 · Getting ready

8 Shopping list

Prices are rough UK prices for small quantities from hobby-electronics shops or online marketplaces. Lots of the list is probably already in the kitchen drawer.

Part	What to look for	Qty	About
N20 gear motor	“N20 gear motor 6 V 300 RPM”, 3 mm D-shaft, ideally with wires already attached	1	£4
Battery box	4 × AA holder with built-in ON/OFF switch and leads	1	£2
AA batteries	Rechargeable NiMH, or alkaline (don’t mix old and new)	4	£5
Push button	Momentary, “normally open”, 12 mm panel type	1	£1
Resettable fuse	PTC “polyfuse”, 500 mA hold current, radial leads	1	£1
Capacitor	100 nF ceramic disc, marked “104”	1	20p
Connectors	Small lever connectors (no soldering needed) and hook-up wire	6	£3
Foam board	A3 sheet, 5 mm thick	1	£3
Stiff plastic	1 mm sheet, e.g. an old ring-binder cover or polypropylene sheet	A4	£1
Wooden strips	10 × 10 mm square dowel for the rails, 450 mm long	2	£2
Lolly sticks	Jumbo craft sticks	6	£2
Bamboo skewers	3 mm, plus narrow straws they spin in freely	4	£1
Split pins	Brass paper fasteners	6	£2
Bottle caps	Four the same size, about 30 mm across	4	free
Rubber bands, cable ties, masking tape	Small band for the belt	—	free
Shallow tray & small tub	Tray ≤ 110 mm wide, 20–30 mm deep; tub about 90 × 50 mm	1 + 1	free
Cress seeds & kitchen paper	One packet does several harvests	1	£1.50

EVERYTHING NEW
≈ £28

USING KITCHEN STUFF
≈ £20

MOST EXPENSIVE
Batteries

9 Tools, and who does what

Tool	Used for	Who
Craft knife, steel ruler, cutting mat	Cutting foam board	Grown-up
Low-temperature glue gun	Almost every joint	Grown-up, or me with a grown-up right there
Bradawl or thick needle	Holes in bottle caps and plastic	Grown-up
Strong scissors	Cutting the plastic guard and knife	Me
Pencil, fine pen, ruler, set square	Marking out	Me
Small screwdriver	Lever connectors, battery box	Me
Sandpaper (fine)	Smoothing the knife edges	Me
Soldering iron (optional)	Only if not using lever connectors	Grown-up

A tidy workspace

- Kitchen table covered with an old newspaper or board.
- Glue gun on its stand, on a heat-proof mat.
- A bowl of cold water nearby, just in case.
- Parts kept in a shoebox between sessions.

Session plan

- **Day 0 (10 min):** plant the cress.
- **Afternoon 1:** templates, chassis, wheels, cutter bar.
- **Afternoon 2:** motor, reel, wiring, ramp, tank.
- **Day 7:** first harvest and sandwiches.

10 Grow the cress first

Cress takes about a week, so plant it **before** building — it will be ready when the machine is.



From seed to harvest. Two days in the dark for strong, straight stems, then a bright windowsill to turn green.

1. **Choose the tray.** No wider than 110 mm and about 20–30 mm deep, washed in hot soapy water. A cut-down food container works.

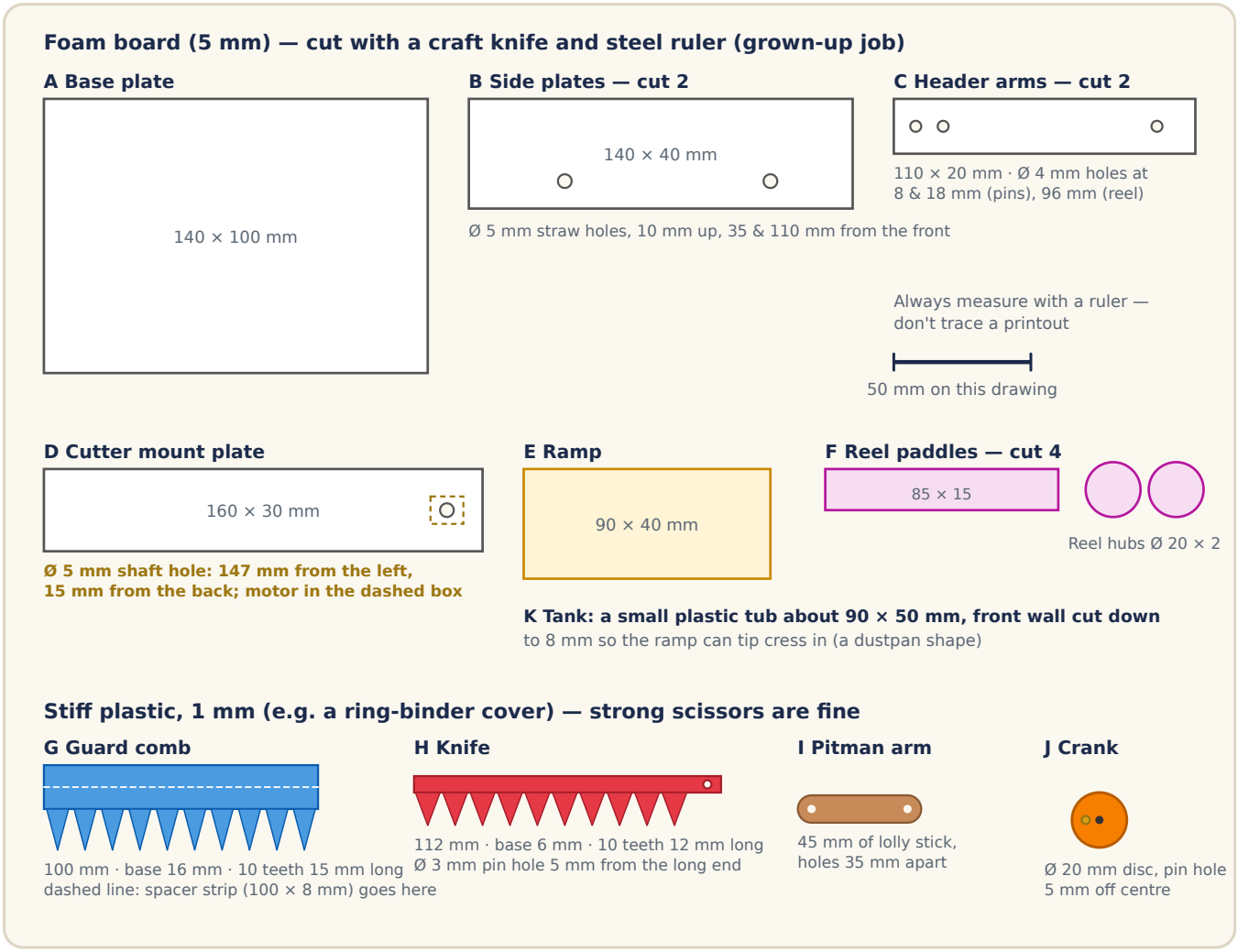
2. **Build the bed.** Fold kitchen paper into a thick pad so its top is about 5 mm below the rim. Soak it, then pour away the extra water.
3. **Sow thickly and evenly.** Sprinkle seeds so they nearly touch, *leaving the first 20 mm at each end bare* so the cutter can start and stop on empty ground.
4. **Days 0–2: dark and damp.** Cover with a plate or put in a cupboard. Mist with a spray bottle if the paper looks pale.
5. **Days 3–7: light.** Uncover on a bright windowsill. Turn the tray every day — cress leans toward the light, and straight stems cut more cleanly.
6. **Harvest at 50–60 mm tall**, when the leaves are fully open and green.

> **SCIENCE BONUS**
>
> Cress leaning toward the window is **phototropism**: the stem grows faster on its shady side, which bends it toward the light. Turning the tray every day keeps it straight — which our knife will like.

Part 4 · Building it

11 Cutting templates

Every part with its size. Mark out with a sharp pencil and a ruler — don’t trace a printout, because printers shrink pages.



Foam-board parts on top (grown-up cuts), plastic cutter parts underneath (scissors). Holes marked with circles; the dashed box on plate D is where the motor sits.

Part	Material	Size	Notes
A Base plate	Foam board	140 × 100 mm	Battery box sits on top
B Side plates × 2	Foam board	140 × 40 mm	Ø 5 mm straw holes 10 mm up, 35 & 110 mm from the front
C Header arms × 2	Foam board	110 × 20 mm	Ø 4 mm holes at 8 & 18 mm (pins) and 96 mm (reel axle)
D Cutter mount plate	Foam board	160 × 30 mm	Ø 5 mm shaft hole 147 mm from the left, 15 mm from the back

Part	Material	Size	Notes
E Ramp	Foam board or card	90 × 40 mm	Glued at about 25°
F Reel paddles × 4, hubs × 2	Stiff card	85 × 15 mm; Ø 20 mm	Paddle tips reach 25 mm from the axle
G Guard comb + spacer	1 mm plastic	100 mm long	Base 16 mm, 10 teeth 15 mm long, 10 mm pitch; spacer strip 100 × 8 mm
H Knife	1 mm plastic	112 mm long	Base 6 mm, 10 teeth 12 mm long; Ø 3 mm pin hole at the long end
I Pitman arm	Lolly stick	45 mm	Ø 3 mm holes 35 mm apart
J Crank	3 layers of plastic	Ø 20 mm	Centre hole for the shaft, pin hole exactly 5 mm off centre
K Tank	Small plastic tub	≈ 90 × 50 mm	Front wall cut down to 8 mm

12 Build 1 — Rails, chassis and wheels

The field

1. **Place the tray** in the middle of a board (an old chopping board or a piece of foam board) about 450 mm long.
2. **Glue a wooden rail on each side**, tight against the tray walls, 120 mm apart centre to centre. The rail tops must be **level with the tray rim**; pack underneath with card if needed.
   grown-up: glue
3. **Leave a run-up.** The rails should stick out about 250 mm past one end of the tray, so the machine can park off the crop.

The chassis

4. **Cut parts A to E** from foam board using the templates. grown-up: craft knife
5. **Make the bearings.** Push a Ø 5 mm hole through each side plate at the marks. Glue a 12 mm piece of straw into each hole. A skewer should spin in the straw freely without wobbling.
   grown-up: glue
6. **Glue the side plates under the base plate**, flush with its two long edges, straw holes towards the bottom.
7. **Axles.** Cut a 150 mm skewer for the back axle and a 170 mm one for the front. Slide them through the straws. The front axle sticks out 20 mm more on the belt side.
8. **Wheels.** Pierce the *exact* centre of four bottle caps. Push them onto the axles, open side facing in, so the wheel centres are 120 mm apart, then glue. Wheels and axles turn together.
   grown-up: bradawl
9. **Rolling test.** Put the chassis on the rails and give it a gentle push. It should roll 30 cm in a straight line without rubbing. If it wanders, one wheel is off-centre — re-glue it.
10. **Handle.** Glue two overlapping lolly sticks to the back edge at about 40°, pointing up and back.

TRAIN LINK

If the wheels keep slipping off the rails, glue a thin card disc slightly bigger than the cap to the *inside* of each wheel. That's a **flange** — the same trick real trains use as a backup.

13 Build 2 — The cutter bar sandwich

This is the most important part of the machine. Take your time: **evenly spaced teeth and a flat, snug knife** matter more than anything else.

1. **Mark the teeth.** On the plastic, draw the guard comb G (100 mm long, 16 mm base) and the knife H (112 mm long, 6 mm base). Mark a dot every 10 mm along each edge with a fine pen.
2. **Cut the teeth.** Snip each V from both sides with strong scissors: guard teeth 15 mm long and 8 mm wide at the root; knife teeth 12 mm long and 9 mm wide. Sand off any whiskers so the edges are crisp and straight.
3. **Pin hole.** Make a Ø 3 mm hole in the knife, 5 mm from the long end. grown-up: bradawl
4. **Spacer.** Cut a 100 × 8 mm strip of the same plastic, add one layer of masking tape on top, and glue it along the *back* 8 mm of the guard base. The spacer is a hair thicker than the knife.
5. **Lay in the knife.** Put the knife on the *front* half of the guard base, in front of the spacer, with its teeth lying over the guard teeth and the long end sticking out 12 mm on the motor side.
6. **Close the sandwich.** Glue mount plate D on top of the spacer, with its front edge level with the front of the guard base. The long end of D, with the shaft hole, goes on the same side as the knife's long end. The plate now overhangs the knife and holds it flat. Don't let any glue reach the knife. grown-up: glue
7. **Slide test.** Push the knife end with a finger. It should slide the full 10 mm with a light push, and must not lift or rattle.

Too loose?

If the knife lifts, glue two small **hold-down** strips of plastic under the front edge of the plate, one near each end, so they just touch the top of the knife.

Too stiff?

Rub a candle over both plastic faces — wax is food-safe and slippery. Check no glue has squeezed into the slot.

14 Build 3 — Motor, crank and pitman arm

1. **Make the crank.** Glue three Ø 20 mm discs of plastic together (or use a flat 20 mm button). Make a snug centre hole for the 3 mm shaft, and a second hole **exactly 5 mm from the centre**. Measure twice — this sets the stroke. grown-up: bradawl
2. **Crank pin.** Push a 12 mm piece of skewer into the off-centre hole and glue it square.
3. **Fit the motor.** From the top, push the N20's shaft down through the shaft hole in plate D until the gold gearbox sits flat. Glue the *gearbox* to the plate and add a cable tie around it. Keep glue off the shaft and the silver motor can. grown-up: glue
4. **Fit the crank** onto the shaft underneath the plate, pin pointing down. Push it on hard, then add a small dab of glue on the flat side of the D-shaft. No glue near the gearbox.

5. **Pitman arm.** Cut 45 mm of lolly stick with two Ø 3 mm holes 35 mm apart. Drop one hole over the crank pin, then glue a bead or a blob of glue onto the *tip* of the pin so the arm can't jump off. Don't glue the arm itself.
6. **Join the arm to the knife** with a brass split pin through both holes. Open the legs gently so the joint swivels freely.
7. **Wiggle test.** Push the knife slowly back and forth by hand. The arm should follow without catching on the plate. If the arm is lower than the knife, pack a small plastic washer under the joint.

> **DON'T FORCE THE MOTOR**
> A tiny gearbox can strip if you twist the shaft hard by hand. If something feels stiff, find what is rubbing rather than forcing it. The first powered test comes after the wiring.

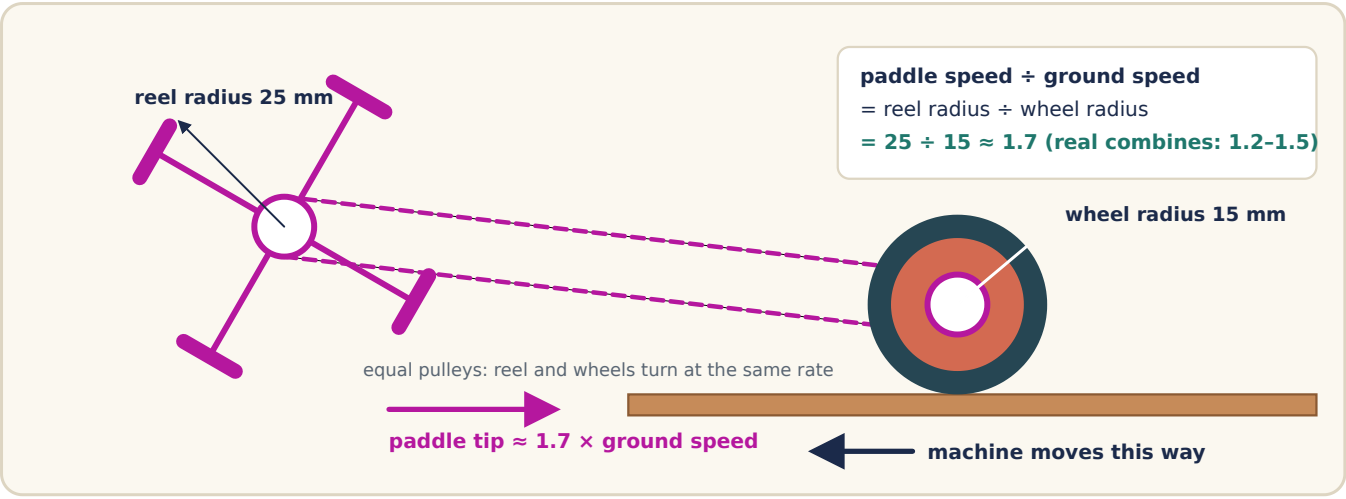
15 Build 4 — Header arms, reel and belt

Attach the header

1. **Glue the cutter unit** (plate D) across the lower front ends of the two header arms C, so the arms are just outside the 90 mm cut width and the motor end hangs out beyond one arm. grown-up: glue
2. **Set the height with a gauge.** Put the chassis on the rails over the tray. Stand a 15 mm block of wood on the cress bed and rest the knife on it. Hold each arm against its side plate and mark through its two pin holes (at 8 and 18 mm).
3. **Pin the arms.** Pierce the marks and fit brass split pins through arm and side plate, both sides. Two pins per arm stop the header swinging.

Reel and belt

4. **Build the reel.** Slide two card hub discs onto a 130 mm skewer, 88 mm apart. Glue four 85 × 15 mm paddles between them at 12, 3, 6 and 9 o'clock. The paddle tips should be 25 mm from the axle.
5. **Mount the reel** through the front holes of both arms so the paddles pass about 5 mm above the knife teeth and a little in front of them. The skewer sticks out 15 mm on the belt side, opposite the motor.
6. **Make two equal pulleys.** Wrap masking tape around the reel skewer end and the front-axle end until each is about 10 mm across. They must be the same size.
7. **Fit the belt.** Stretch a small rubber band between the two pulleys, not crossed. Push the machine forward by hand: the *bottom* paddles must move *backward*, towards the knife. If they go the wrong way, cross the band into a figure of eight.



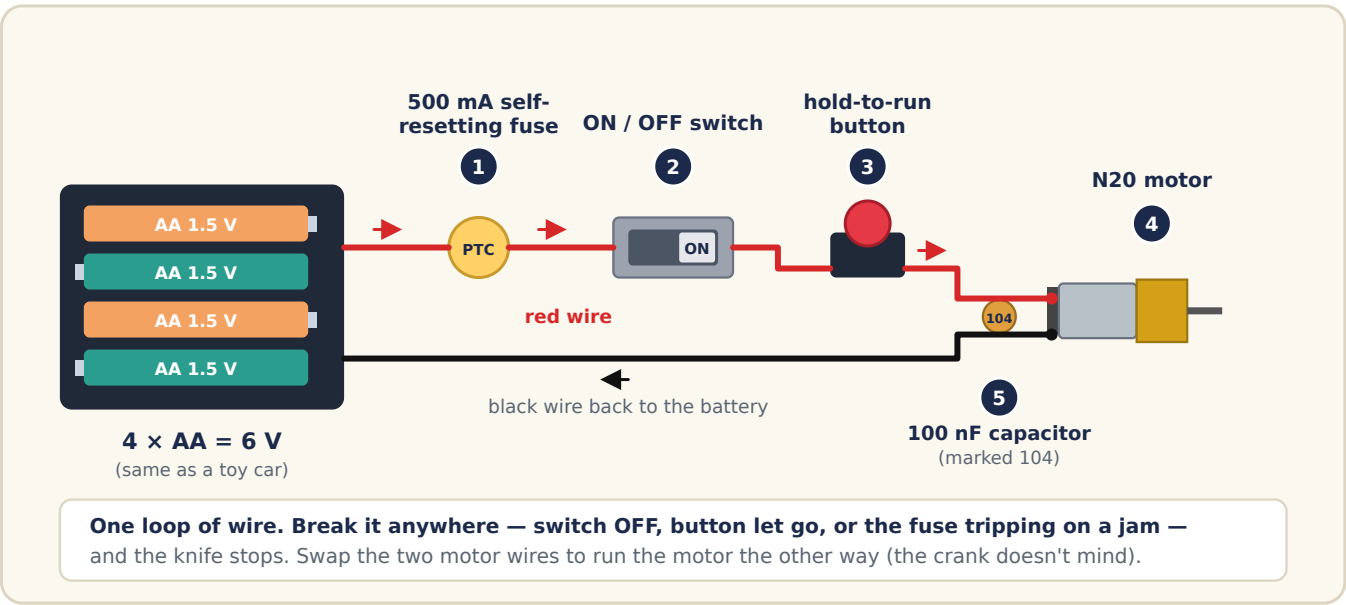
Equal pulleys mean the reel turns once for every turn of the wheels. The reel is bigger than the wheel, so its paddle tips move about 1.7 times faster than the machine — just enough to lean the cress back onto the knife.

A BUILT-IN CLUTCH

If the reel catches on something, the rubber band just slips. Nothing breaks, and the knife motor isn't affected.

16 Build 5 — Wiring

The whole circuit is **one loop**: power leaves the battery on the red wire, passes through three safety parts and the motor, and comes back on the black wire.



Red is the + side, black is the − side. Numbers match the steps below. If the battery box has a built-in switch, that is part 2.

1. **Batteries OUT** of the box while wiring. Always.
2. **Capacitor.** Connect the 100 nF "104" capacitor across the two motor terminals (or across the two motor wires, close to the motor). It stops the motor crackling on radios and TVs. Either way round is fine. grown-up if soldering
3. **Fuse first.** Connect the battery red wire to one leg of the resettable fuse with a lever connector.

4. **Switch, then button.** From the fuse, run red wire to the ON/OFF switch (skip if it’s built into the box), then to the hold-to-run button, then to the motor’s red wire.
5. **Black wire** from the battery straight to the motor’s black wire.
6. **Tidy up.** Hot-glue the button into a hole in the top of the handle. Tape the wires along the handle and chassis, and use cable ties to keep them well away from the crank.
7. **First spin.** Lift the machine off the rails. Switch ON, then press the button for two seconds. The knife should blur side to side and stop the moment you let go. It doesn’t matter which way the motor turns.

1 · Fuse
If the knife jams, the motor draws more current. The PTC fuse warms up and **switches itself off**. It resets after cooling for a minute.

2 · ON/OFF switch
The master switch. OFF whenever the machine isn’t on the rails.

3 · Hold-to-run
Only runs while pressed. Let go and it stops — like a lawnmower’s safety bar.

17 Build 6 — Ramp, tank and first harvest

Ramp and tank

1. **Cut the tank.** Cut the front wall of a small plastic tub down to 8 mm high, making a dustpan shape. (grown-up: scissors or knife)
2. **Fit the tank** under the front of the base plate, just behind the cutter unit, with tape or a dab of glue. Its bottom must clear the cut stubble by at least 5 mm.
3. **Fit the ramp.** Glue ramp E from the back edge of plate D up to the low front wall of the tank, at about 25°. A gentle slope matters: cress won’t climb a steep one.

First harvest

4. **Safety check** with a grown-up, using the checklist on the safety plan.
5. **Practice run** on a spare tray of long grass, or a few cress stems, before the real crop.
6. **Park** the machine on the run-up, knife just before the crop. Switch ON.
7. **Hold the button and push slowly** — about 20 mm a second, so a 20 cm tray takes 10 seconds. Keep one steady speed.
8. **At the far end**, let go of the button, switch OFF, and take the batteries out before touching anything near the knife.
9. **Empty the tank** into a sieve, rinse the cress under the cold tap, and pat it dry.

EGG & CRESS SANDWICHES (FOR 2)
Hard-boil 2 eggs (10 minutes), cool in cold water, peel and mash with a spoon of mayonnaise and a pinch of salt and pepper. Spread on buttered bread, scatter a big handful of freshly harvested cress on top, close, and cut into triangles. Farm to table in 15 minutes.

Part 5 · Fixing and upgrading

18 Troubleshooting

Something not working is part of the project. Change **one thing at a time** and test again.

What you see	Why	What to do
Nothing happens	No power reaching the motor	Check batteries are in the right way, the switch is ON, the lever connectors are closed, and the fuse isn't still warm
Motor hums, knife doesn't move	Jammed or the sandwich is too tight	Batteries out, clear the jam, wax the knife, check the arm isn't rubbing the plate
Stops, then works again a minute later	The fuse is doing its job on a jam	Find the stiff spot; push more slowly through thick patches
Stems bend over instead of cutting	Knife lifting, or pushing too fast	Add hold-down strips; push more slowly; check knife teeth really overlap the guard teeth
Ragged, torn cut	Rounded or whiskery tooth edges	Recut the knife with sharp scissors; sand edges crisp; try the 3D-printed cutter (Upgrade 6)
Crank arm jumps off	Crank pin too short, or no stop on its tip	Longer pin with a bead glued on the tip
Reel doesn't turn	Belt slipping	Smaller rubber band; add another layer of tape to both pulleys
Reel throws cress forward	Reel turning the wrong way	Cross the belt into a figure of eight
Cress piles up at the bottom of the ramp	Ramp too steep or too far from the reel	Lower the ramp angle; move the reel back so paddles reach the ramp; or try the conveyor (Upgrade 7)
Knife scrapes the paper	Bed too high or lumpy	Flatten the bed; raise the header one row (Upgrade 1)
Machine wanders off the rails	Off-centre wheel, or axles not square	Re-centre wheels; add card flanges inside the wheels
Radio or TV crackles	Electrical noise from the motor brushes	Check the 100 nF capacitor is connected right at the motor

19 Upgrades — overview

Once the basic machine works, here is how to make it better. Each upgrade is on its own page with its own steps. Do them in any order, one at a time, and test after each.

#	Upgrade	Difficulty	Extra cost	You learn about
1	**Adjustable cutting height**	Easy	Free	Measuring, fits, gauges
2	**Speed dial**	Medium	£3	PWM electronics
3	**Self-driving**	Medium	£5	Gear ratios, speed

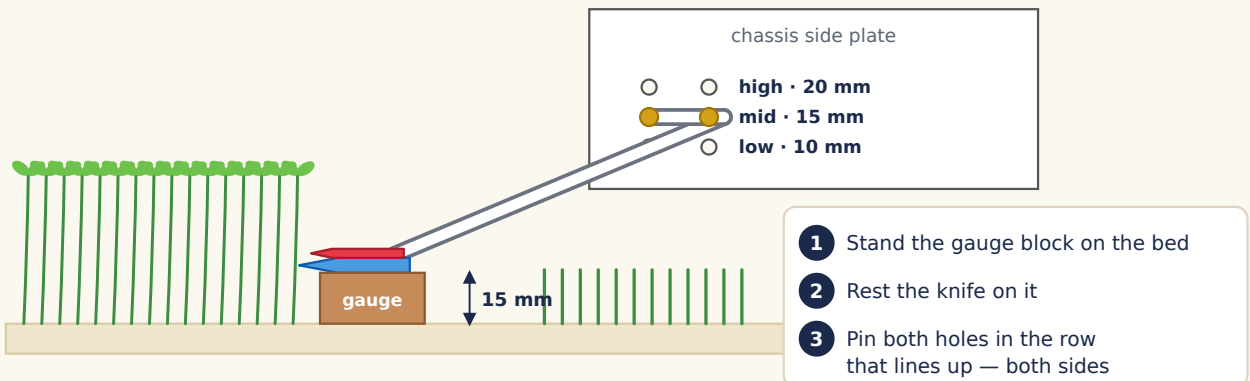
#	Upgrade	Difficulty	Extra cost	You learn about
4	**Automatic end-of-row stop**	Medium	£1	Safety interlocks
5	**Wider header**	Medium	£1	Load, torque, scaling
6	**3D-printed cutter bar**	Advanced	Filament	CAD, tolerances
7	**Conveyor-belt feeder**	Advanced	Free	Belts, rollers, timing

THE UPGRADE RULE

Batteries out before changing anything. After every upgrade, run through the first-spin test and the safety checklist again before cutting cress.

20 Upgrade 1 — Adjustable cutting height

Different crops need different heights. Small cress wants a low cut; a lumpy bed needs a higher one. Real combines let the driver raise and lower the header from the cab; ours uses three rows of pin holes.



Small cress (day 5): 10 mm · Normal: 15 mm · Lumpy bed or knife scraping: 20 mm

Two pins per arm keep the header level. Moving both pins up or down one row lifts or lowers the whole cutter by 5 mm without tilting it.

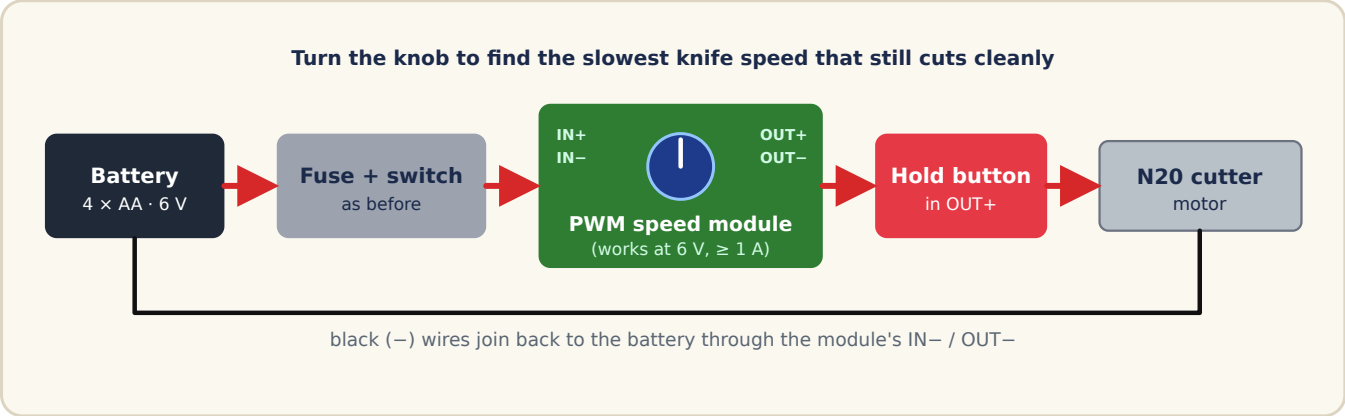
1. **Mark the rows.** On each side plate, around the two existing pin holes, mark one extra pair of holes 5 mm above and another 5 mm below. That’s three rows, two holes in each, 10 mm apart.
2. **Make the holes** Ø 4 mm. Put a small square of tape over the foam first so the holes don’t tear. grown-up: bradawl
3. **Make gauge blocks** 10, 15 and 20 mm tall from scrap wood, or folded card, and write the size on each.
4. **To change height:** batteries out, stand the gauge on the bed, pull both pins on each side, rest the knife on the gauge, and push the pins into the row that lines up. Use **the same row on both sides**.
5. **Check the ramp and reel** still clear everything, then do a first-spin test.

Row	Cut height	Use when
Low	10 mm	Young cress (day 5), shorter stems

Row	Cut height	Use when
Mid	15 mm	Normal harvest
High	20 mm	Lumpy bed, or the knife scrapes the paper

21 Upgrade 2 — Speed dial

A **PWM speed controller** is a little circuit board with a knob. It switches the power on and off hundreds of times a second. More “on” time means a faster motor. It lets you find the best knife speed by experiment, instead of buying different motors.



The module goes after the fuse and switch. The hold-to-run button stays in the motor’s red wire, so letting go still stops the knife instantly.

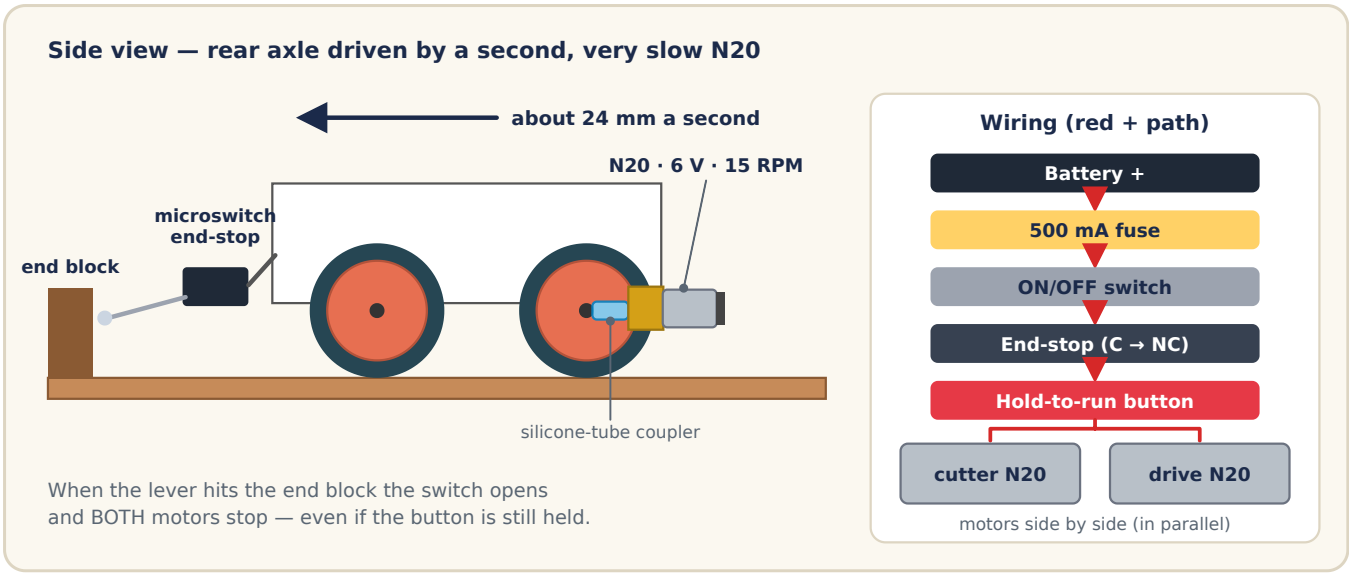
1. **Buy** a small “PWM DC motor speed controller” board with a knob. Check the listing says it works from **6 V or less** and handles at least **1 A**.
2. **Batteries out.** Disconnect the red wire between the switch and the button.
3. **Input:** red wire from the switch to `IN+`; a new black wire from the battery’s black to `IN−`.
4. **Output:** `OUT+` to the hold-to-run button, and the button to the motor red. Motor black goes to `OUT−` instead of straight to the battery.
5. **Mount** the board on the base plate with sticky pads, knob towards the handle. Keep it dry: no cress water near it.
6. **Experiment.** Start with the knob at halfway. Cut a test strip, then try faster and slower. Write down the slowest setting that still gives a clean cut.

WHY NOT JUST GO SLOW?

Turning the knob down lowers the knife’s power as well as its speed, so at very slow settings a thick patch can stall it, and the fuse will trip. Fewer cuts a second also means you must push more slowly. Use the simulator to see why.

22 Upgrades 3 & 4 — Self-driving with automatic stop

Pushing at a perfectly steady speed is hard. A second, **very slow** N20 on the back axle drives the machine along at exactly the same speed every time. The automatic stop switch makes sure it can never drive off the end.



Left: the drive motor on the back axle and the end-stop switch at the front. Right: the order of parts along the red wire. Both motors sit after the hold-to-run button.

Upgrade 3 — Drive motor

1. **Buy** a second N20, **6 V, 15 RPM**, and 20 mm of silicone tube with a 3 mm hole.
2. **Work out the speed:** wheel circumference = π × 30 mm ≈ 94 mm. 15 turns a minute × 94 mm = 1,410 mm a minute, or **about 24 mm a second**. With 10 knife cuts a second, that's 2.4 mm between cuts — well within the golden rule.
3. **Lengthen the back axle** so it sticks out 15 mm on one side. Push the silicone tube half onto the axle and half onto the motor shaft.
4. **Mount the motor** on a foam-board block glued to the side plate, so the shaft lines up exactly with the axle. The tube soaks up small misalignment. grown-up: glue
5. **Wire it in parallel** with the cutter motor: its red to the cutter motor's red, its black to the cutter motor's black — after the button. Add its own 100 nF capacitor.
6. **Check the direction.** Press the button with the machine lifted: the wheels must turn so it would drive *forward*. If not, swap the drive motor's two wires.

Upgrade 4 — End-of-row stop

7. **Buy** a lever microswitch with three terminals: C, NO and NC.
8. **Wire C and NC** into the red wire, between the ON/OFF switch and the hold-to-run button. Not NO! NC means "normally closed": power flows until the lever is pressed.
9. **Mount it** at the front of the chassis above one rail, lever pointing forward and down. Glue a wooden **end block** on that rail just past the end of the tray.
10. **Test it before any cress:** hold the button so both motors run, then press the lever with a finger. **Everything must stop.** If not, re-check you used NC.

A REAL SAFETY INTERLOCK

This is the same idea as the switch that stops a lift moving while its doors are open. The machine is safe even if someone forgets to let go of the button.

23 Upgrades 5 & 6 — Wider header and 3D-printed cutter

Upgrade 5 — Wider header

For a wider tray, everything across the machine gets longer by the same amount. Keep the tooth pitch at 10 mm and the stroke at 10 mm.

Part	Standard	For a tray W mm wide
Rail spacing (centres)	120 mm	W + 10 mm
Guard comb / number of teeth	100 mm / 10	W − 10 mm / (W − 10) ÷ 10
Knife	112 mm	guard + 12 mm
Plate D	160 mm	guard + 60 mm
Base plate, axles, reel paddles	100 / 150 / 85 mm	each + (W − 110) mm

1. **Mind the load.** Every extra tooth adds rubbing, so the fuse will trip sooner. Up to about 150 mm wide, the 300 RPM motor should cope; test by running with no crop and feeling if the motor gets warm.
2. **Wider than 150 mm?** Either build **two cutter units**, one from each side, each with its own motor and knife meeting in the middle, or use a stronger **100 RPM** N20 (about three times the turning force).
3. **With a 100 RPM motor** there are only 3.3 cuts a second, so the golden rule limits you to **16 mm a second** push speed.

Upgrade 6 — 3D-printed cutter bar

If there’s a 3D printer available (like the Ender-3 used for the PlaneForge aircraft), printed parts are straighter and more even than hand-cut plastic — and a better cut comes from perfectly spaced teeth.

4. **Design** in Tinkercad (free, in a web browser): guard comb, knife, and a cutter body that includes the spacer and the hold-down lip in one piece.
5. **Sizes:** knife 1.0 mm thick; slot for it 1.3 mm; guard 1.2 mm; same teeth as the templates.
6. **Crank with a D-hole:** the N20 shaft is 3 mm with a flat at 2.5 mm. Print a test hole first, then adjust by 0.1 mm until it pushes on snugly without glue.
7. **Print** flat on the bed in **PETG** (tougher and slipperier than PLA), 100% infill, 0.2 mm layers. Clean the first-layer “elephant’s foot” off the knife edges with fine sandpaper.
8. **Swap it in** using the same motor, arm and split pin, then re-check the slide test.

24 Upgrade 7 — Conveyor-belt feeder

Real combines don’t rely on a slope — they use a **conveyor** (the feeder house) to carry the crop up. Replacing the ramp with a small moving belt means cress never piles up at the bottom.

What it is

A loop of 20 mm ribbon or a cut-open rubber glove cuff, running over two skewer rollers: one just behind the knife, one above the tank’s low front wall. Matchstick cleats glued across it every 15 mm grab the cress.

What drives it

A second small rubber band from the reel’s axle to the top roller. Like the reel, it only moves when the machine moves — so it needs no extra motor or wiring.

1. **Remove the ramp** (a warm butter knife run under hot glue releases it). grown-up
2. **Rollers.** Cut two 100 mm skewers. Wrap each with tape to 8 mm across in the middle, so the belt tracks centrally.
3. **Bearings.** Glue short straws to the insides of the header arms: bottom roller just behind plate D, top roller above the tank lip. The belt should slope at about 30°.
4. **Belt.** Loop the ribbon around both rollers and join the ends with a strip of tape on the *outside*, slightly tight. Glue on matchstick cleats every 15 mm.
5. **Drive.** Make a tape pulley on the reel axle (outside the arm) and another on the top roller, and join them with a small rubber band.
6. **Direction check:** push the machine forward by hand. The *top* of the belt must move *up and back* towards the tank. If not, cross the band.

KEEP FINGERS CLEAR

The conveyor has no motor of its own and stops when the machine stops, but rollers can still pinch. The batteries-out rule applies before clearing it.

Part 6 · Safety and summary

25 The safety plan

For Mum: every risk we could think of, and what we do about each one.

Risk	What we do about it
Craft-knife cuts	A grown-up does all knife cutting on a mat with a steel ruler; the blade is retracted and put away straight after.
Hot-glue burns	Low-temperature glue gun, used by or right next to a grown-up; kept on its stand; bowl of cold water nearby.
Fingers in the cutter	Plastic teeth, not metal; scissor action, not a spinning blade; hold-to-run button; guards stick out further than the knife; **batteries out before hands go near the knife**.
Electric shock	Not possible: 6 V from AA batteries is far too low to be felt. No mains electricity anywhere in the project.
Battery short circuit	Batteries live in a proper box; the fuse is the first thing after the battery; batteries out while wiring and when the machine is put away.
Motor jam or overheating	Self-resetting 500 mA fuse; let go of the button if it stalls; the belt slips instead of breaking.
Small parts	Split pins, beads and caps kept in a closed box, away from younger children and pets.
Food hygiene	Washed tray; clean hands; only candle wax as a lubricant; electrics above and behind the crop; cress rinsed before eating; any cress touched by glue thrown away.
Water and mess	Tray on a board with a tea towel underneath; misting bottle instead of a jug; electronics kept dry.

Rules card (stick it on the handle)

1. **Batteries OUT** before fingers go near the knife.
2. Only press the button when the machine is **on the rails**.
3. **Both hands on the handle** while it runs.
4. Grown-ups do **knives and hot glue**.
5. Smells hot or stops? **Let go, switch OFF, batteries out.**

Before every run — checklist

- ☐ Knife slides freely, no cracked teeth
- ☐ Crank pin has its stop on the tip
- ☐ Wires tied clear of the crank and wheels
- ☐ Button stops the knife the moment it’s released
- ☐ End-stop switch tested (if fitted)
- ☐ Tray on its board, towel underneath

26 What I’ll learn

This isn’t just a toy. Nearly every school science and maths topic turns up somewhere in it.

Biology

How seeds germinate, why seedlings grown in the dark are tall and pale, phototropism, and where food actually comes from.

Maths

Measuring to the millimetre, ratios (25 ÷ 15), speeds and rates (cuts per second), circumference (π × diameter), and working out a budget.

Physics & engineering

Turning rotation into sliding with a crank; gearboxes trading speed for force; how shearing works; electric circuits, fuses and switches; testing and fixing.

Life skills

Following a plan, working safely, being patient for a week, fixing things that don't work first time — and making lunch for the family.

> *The engineering design process: imagine, plan, build, test, improve.*
>
> — this project does all five, then the upgrades do them again

27 Jargon buster

Word	What it means
N20 motor	A standard tiny electric motor (12 × 10 mm) sold with a gearbox attached
RPM	Revolutions per minute — how many whole turns the shaft makes each minute
Gearbox	Gears that make the output slower but stronger
D-shaft	A motor shaft with one flat side, so things fixed to it can't slip round
Torque	Turning force — how hard the motor can twist
Stall	When the motor is stuck and can't turn; it then draws the most current
Crank	A disc or arm with a pin off-centre, turning round and round
Pitman arm	The rod that turns the crank's circling into the knife's back-and-forth
Reciprocating	Moving back and forth in a straight line
Stroke	How far the knife travels each way (10 mm)
Pitch	The distance from one tooth to the next (10 mm)
Header	The whole front of a combine: reel, cutter bar and feeder
Shear	Cutting by sliding two edges past each other, like scissors
PTC fuse	A fuse that turns itself off when too much current flows, then resets when it cools
Capacitor	A tiny part that soaks up electrical crackle from the motor
PWM	Pulse-width modulation: switching power on and off very fast to control speed
NC / NO	Normally closed / normally open — whether a switch lets power through before you press it
In parallel	Two parts wired side by side so each gets the full battery voltage

28 Cheat sheet — so, Mum…?

- **What:** a cress combine harvester, about 24 cm long, rolling on rails beside a shallow tray of home-grown cress.
- **How it cuts:** a 6 V, 300 RPM **N20 gear motor** turns a crank; a pitman arm slides a plastic knife 10 mm over a plastic guard comb — **10 scissor cuts a second**.
- **How it collects:** a wheel-driven reel sweeps cress onto the knife; a gentle ramp leads into a dustpan-shaped tank.
- **The golden rule:** move less than 5 mm between cuts — push at about 20 mm a second.
- **Safety:** 6 V batteries, plastic knife, fuse → switch → hold-to-run button, grown-up does knives and glue, batteries out before touching the cutter.
- **Cost and time:** about £20, two afternoons of building, one week of growing.
- **Upgrades:** adjustable height, speed dial, self-driving with an automatic stop, wider header, 3D-printed cutter, conveyor feeder.

SO, MUM… CAN WE?

It’s safe, cheap, tidy and planned down to the last millimetre. I’ll learn about plants, maths, machines and circuits — and at the end of the week, you get an egg & cress sandwich made from a crop I grew and harvested with a machine I built. Everything is also in the PDF report, so you can read it properly first.

A Appendix — parts checklist and pre-run sign-off

Parts checklist

- [ ] N20 gear motor, 6 V, 300 RPM
- [ ] 4 × AA battery box with switch
- [ ] 4 AA batteries
- [ ] Momentary push button (normally open)
- [ ] Resettable PTC fuse, 500 mA
- [ ] 100 nF ceramic capacitor (“104”)
- [ ] Lever connectors and hook-up wire
- [ ] A3 foam board, 5 mm
- [ ] A4 stiff plastic, 1 mm
- [ ] 2 × wooden strips 10 × 10 × 450 mm
- [ ] 6 jumbo lolly sticks
- [ ] 4 bamboo skewers + narrow straws
- [ ] 6 brass split pins
- [ ] 4 matching bottle caps
- [ ] Rubber bands, cable ties, masking tape
- [ ] Shallow tray (≤ 110 mm wide) and a small tub
- [ ] Cress seeds and kitchen paper
- [ ] Board for the field, candle for wax

Pre-run sign-off (tick every item before each harvest)

- [ ] Knife slides freely; no cracked or bent teeth
- [ ] Crank pin has its stop glued on the tip
- [ ] Split pins on the arms and knife are secure
- [ ] Wires tied clear of the crank, belt and wheels
- [ ] Releasing the button stops the knife at once
- [ ] End-stop switch stops both motors (if fitted)
- [ ] Machine runs straight on the rails with no crop
- [ ] Tray on its board, towel underneath
- [ ] Hands washed; tray clean; cress will be rinsed
- [ ] Everyone knows: batteries out before touching the knife

Builder

Grown-up